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# Energy consumption parameter analysis of industrial robots using design of experiment methodology

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**ABSTRACT**

During the last decade, industrial robot installations have increased greatly, contributing to higher energy consumption at the process level. To align with Industry 4.0 objectives to reduce energy consumption, further Industrial Internet of Things, and advance Smart Manufacturing, this paper explores energy consumption optimisation of process-level industrial robots through Design and Analysis of Experiment methodologies. The operating parameters of a Kawasaki ZZX130L, 6 DOF model industrial robot, investigated herein are speed, acceleration, payload, and temperature using a linear factorial experiment analysis. It is this paper's goal to determine which of the explored parameters contribute most to energy consumption. Statistical analysis of data was conducted using Minitab 19, an industry standard tool, and it was found that linear speed and acceleration contributed to nearly 95% of energy consumption in any of the first three joints of the Kawasaki robot, with the other factors, payload and temperature, contributing the rest. It remains important to explore other factors contributing to EC and seek to minimise them in any way possible.

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**KEYWORDS**
Energy consumption; design of experiments; process level; industrial robotics; parameter optimisation

## 1 Introduction

According to the Industrial Federation of Robotics (2019), annual sales of industrial robots (IR) has risen every year for the past decade, with well over 400,000 new units expected to be installed in 2019 alone and continued increases in sales predicted for the future. The automotive and electronics industries represent 55% of all installations (IFR 2019), where they consume 8% of total energy in a given production process (Paryanto, Bornschlegl, and Franke 2015). For this reason, the goal of reducing energy consumption (EC) in IR is not trivial with potential savings aligning well with Industry 4.0 objectives and contributing heavily to eco-factories of the future. With the rise of Industry 4.0, the automation of the manufacturing industry is being rapidly developed with goals to achieve higher energy and production efficiency through data sharing and cloud computation using Industrial Internet of Things (IIOT) methodologies (Wang et al. 2016). By connecting IR to the cloud, data such as energy usage at a production level, total idle times, and other useful information can be gathered and used to inform future manufacturing process decisions.

As found by Tantawi, Fidan, and Tantawy (2019), large robotics manufacturers are rolling out their own IIOT platforms with the number of connected robots ranging from 7,000 to more than one million devices, which includes both robots and sensors, as of 2018. Based on the rapid growth of Smart Manufacturing (SM) practices in alignment with Industry 4.0 objectives, the deployment of such practices to small- and medium-sized manufacturers is projected to have an energy cost savings of 25 USD billion (Tantawi, Fidan, and Tantawy 2019). Our research is focused on techniques to reduce energy usage to support sustainable manufacturing practices and processes.

Research in the field to reduce IR EC has been historically focused on motion planning as in the research by Mohammed et al. (2014), and scheduling as in Chemnitz, Schreck, and Krüger (2011), though most research is actually a subset of optimising operating parameters such as acceleration, speed, and load as in the research by Smetnova (2010); Nevmerzhitskiy et al. (2019); and Eggers et al. (2018). Our research focuses on understanding which operating parameters most contribute to EC, so we will briefly address both motion planning and scheduling before discussing operating parameter optimisation as a whole.

### 1.1 Motion planning

First and foremost, motion planning typically focuses on improving motion profiles, smoothing out movements to minimise acceleration and optimise energy consumption. Of the three focuses, motion planning tends to have the largest energy savings (Paryanto, Bornschlegl, and Franke 2015). For processes such as welding, motion profiles cannot be modified to save energy as certain processes must be performed at set speeds and accelerations, so this approach is better for other processes and for implementation in the planning stages of new factories. In general, although it may be possible to implement this type of energy savings in existing factories, most methods do not account for subsequent changes in cycle times.

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The work of Mohammed et al. (2014), proposes a simulated analysis of energy consumption based on arm joint location within a robot's workspace based on Denavit-Hartenberg notation. This work enabled the researchers to find paths and poses within the robot's workspace that purposefully avoided regions of high energy usage. Overall, this work was based on information such as payload, velocity, and acceleration, though the latter two were path-type (straight or square) dependent. Naturally, the researchers found that with payload EC increased, though depending on the path-type, this increase in payload resulted in a variation in EC from 2 to 10 Joules.

In the experiment by Paes et al. (2014), they focus on reducing energy consumption through efficient trajectory planning by intelligent programming. This most-efficient trajectory is found through having the robot perform a determined excitation routine, which is used to identify the dynamic model parameters. While this method has been shown to reduce energy consumption up to 5% and reduce cycle times by up to 4%, the excitation routine must be chosen per robot per process type. The strength in this method is application to a line of robots that are performing the same tasks, and that the method can be easily applied in situ.

Finding and programming the most energy-efficient path for any number of processes or robots would almost certainly be too tedious to implement. Yin, Ji, and Wang (2019) approached this problem using a machine learning approach to allow the industrial robot to train itself into the most efficient trajectories. The proposed method allows for great universal application and further understanding of the industrial robot's energy usage over its working lifetime. Due to the nature of the work and current regulations, having a self-teaching robot create its own work path is not yet completely viable, but with improvements to overall safety systems, such as those found in collaborative robotics, this method could be incredibly useful.

### 1.2 Scheduling

Next, scheduling is a subset of motion planning that aims to minimise idle time, taking advantage of optimising the motions between the principal actions. Chemnitz, Schreck, and Krüger (2011) have shown that there is a strong robot model-dependence on whether idling paired with fast movement or no idling paired with slow movement is ideal. This research suggests that either age of robot or manufacturer can heavily contribute to robot energy efficiency. Continued monitoring of industrial robots currently in use would likely be able to identify an industrial robot's breakdown overtime or indicate when it needs to be overhauled.

In the theoretical investigation conducted by Pastras, Fysikopoulos, and Chryssolouris (2019), it was found that motion planning can be used in conjunction with scheduling to reduce EC and maintain cycle times. This method takes advantage of regions of low energy usage and combines it with modifying movement time duration like the above study. By combining motion planning with scheduling, or having any combination of optimisation methods, energy reduction is easily attainable and applied.

Further theoretical investigations by (Bryan, Grenwalt, and Stienecker 2010) suggest that there is no need to hard code a single most-energy-efficient programme, when it is possible to have the robot teach itself to adapt to paths that best minimise EC, while maintaining cycle time constraints.

The scheduling method of EC reduction is rather robust and can easily be implemented in existing production processes as found by Meike and Ribickis (2011), as it does not modify cycle times, which means factories would not have to modify any other lines to comply with lower energy usage.

### 1.3 Parameter optimisation

Finally, research into optimising operating parameters tends to focus on either one or two variables with the assumption that they are largely independent of the effects of other variables. For example, though Smetnova (2010) found that, regardless of payload, there existed an EC minima at approximately 30% of maximum acceleration and between 30% and 50% of maximum linear speed (also verified by Liu et al. 2018), it is not necessarily true that selecting both operating parameters to be those values will lead to the optimal EC condition (Montgomery 2017). This is due to potential interactions between variables.

Other than the common research on speed, acceleration, and load, a fourth variable, temperature, is often investigated due to its role in both electrically and mechanically related thermal effects. For example, Pirzadeh, Mohammadion, and Lavasani (2019) explored temperature effects on a robot's repeatability. Nevmerzhitskiy et al. (2019) explored temperature, load, and speed's effects on the friction function, which, along with the previous example, could also lead to interesting effects on EC as a whole.

Eggers et al. (2018) explored the relationship between travel time and EC, proposing an energy model to simulate the energy usage of IR, and in a later paper, Eggers et al. (2019) investigated temperature effects on EC, using the results to increase their model's accuracy. The temperature was taken on the outside of the motor casing and taken as the relative temperature of the internal gearbox, leading to the conclusion that higher temperatures reduced mechanical losses, increasing energy efficiency.

Though the literature is full of methods to reduce EC, few if any propose which factors affect EC the most. As a result, the Design of Experiments and Analysis methodology, as written in Montgomery (2017), is a systematic method to understand how input parameters and their interactions affect output parameters. This means this particular method is both robust and widely applicable from planning to understanding existing factories for process-level events. For our experiment, our inputs and output are further discussed in Section 3.

## 2 Methodology

We chose to investigate EC with the Kawasaki ZZX130L, which has a top linear speed of 2500 mm/s, a maximum payload capacity of 130 kg, and 6 degrees of freedom (DOF) (Kawasaki Heavy Industries, 2015). The acceleration of the robot is limited by its slowest joint, in this case, those would be joints 1, 2, and 3, though no manufacturer-specified value exists. 6-DOF IR are

used for a wide variety of applications, though our model was manufactured specifically for various types of welding and assembly, and more generally, pick and placing. To that end, we chose to simulate a generalised pick and place motion with a static payload to understand EC at the process level. The joint motions were isolated from one another to better understand each joint on their own, which was enabled by programming the robot in joint mode. The last three joints were held motionless.

According to Liu et al. (2018), the first three joints, as counted from the bottom, are responsible for about 80% of all energy consumed for an IR. As a result, we chose to measure only the EC of those first three joints (hereafter referred to J1 (arm rotation), J2 (arm out-in), and J3 (arm up-down)). Our robot's controller has a built-in feature to measure individual joint motor currents, which we used in our calculation of joint EC. Power was found using conventional methods for calculating 3-phase motor power and summed to find total power over a single cycle.

Our experimental setup, as seen in Figure 1, is as follows: individual motor currents are collected from the robot's controller by direct, wired connection to the computer. We are able to manipulate the robot's payload between experiments using weights, each approximately 20 kg, bolted on using a 5 kg mounting plate. Temperature data were acquired by attaching adhesive K-type thermocouples to the outside of J1, 2 and 3's motor casings. The assumption was made that the heat conducted into the outside of the motor casings would be relative to the internal robot temperature, with some hysteresis, due to gear boxes generating relatively little heat on their own through friction because of the use of high-quality grease. It was assumed that most of the heat affecting the robot would be generated from the motors and conducted into the robot, hence the choice of motor casing temperature readings. An additional thermocouple was attached to the outside of J2's gearbox (J2 GB) in an effort to verify the assumption that the gearbox would not heat up greatly due to friction, and a final thermocouple was used to record ambient room temperature. The temperatures of each joint and gearbox can be seen in Figure 2, which verifies that gearbox temperature does not increase greatly over the course of the experiment, when compared with the motor casings which heat up more dramatically. This temperature data was read into a custom LabVIEW program, which outputted a comma-separated value file. That data was later imported into Minitab for statistical analysis.

EC data were acquired while the robot was in both low and high-temperature states. In order to achieve the high-temperature value, which was read from J1 for convenience and consistency, we ran the experimental sequence continuously at 100% speed and acceleration values with the 65 kg load attached as the warm up sequence. As seen in Figure 2, further discussed in Section 5, the warm up sequence took a little more than 2 hours to achieve a J1 temperature of 40°C.

## 3 Design of experiments (DOE)

Experiments were set up through the concept of 24 full factorial DOE, with each individual experiment representing an independent combination of factors (Montgomery 2017). The independent parameters, further discussed in Section 3.1, of linear speed, acceleration, payload, and temperature were selected to understand the EC response variable. The factor levels and specifications are shown in Table 1.

### 3.1 Parameter selection

Speed, acceleration, and payload are common variables found in many types of mechanical experiments, usually due to the ease of testing and due to their relevance in energy consumption (Garcia, Bittencourt, and Villani 2018). For the Kawasaki ZZX130L, these variables are very easy to manipulate. Speed is defined as the linear speed the robot can achieve, while acceleration and payload have their common definitions. In the robot's controller, speed and acceleration can be modified as percentage values. We chose temperature as a variable to explore due to existing research by Eggers et al. (2018) which improves existing energy modelling accuracy with the addition of temperature; this will enable us to understand the exact contributions temperature has on EC. In both Eggers et al. (2018) and Eggers et al. (2019), the author suggests a strong correlation with temperature effects on energy consumption and proposes that a change from 20°C to 70°C can increase energy consumed by as much as 1 kW. Additionally, the robot will be simulating a pick and place

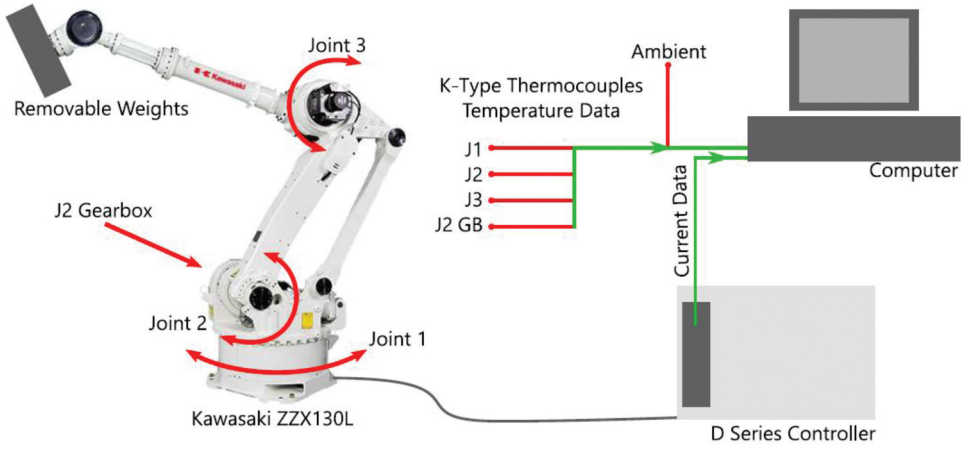



Figure 1. Experimental set up for measuring temperatures (joints 1–3, joint 2 gearbox and ambient) and getting motor current data from IR controller.

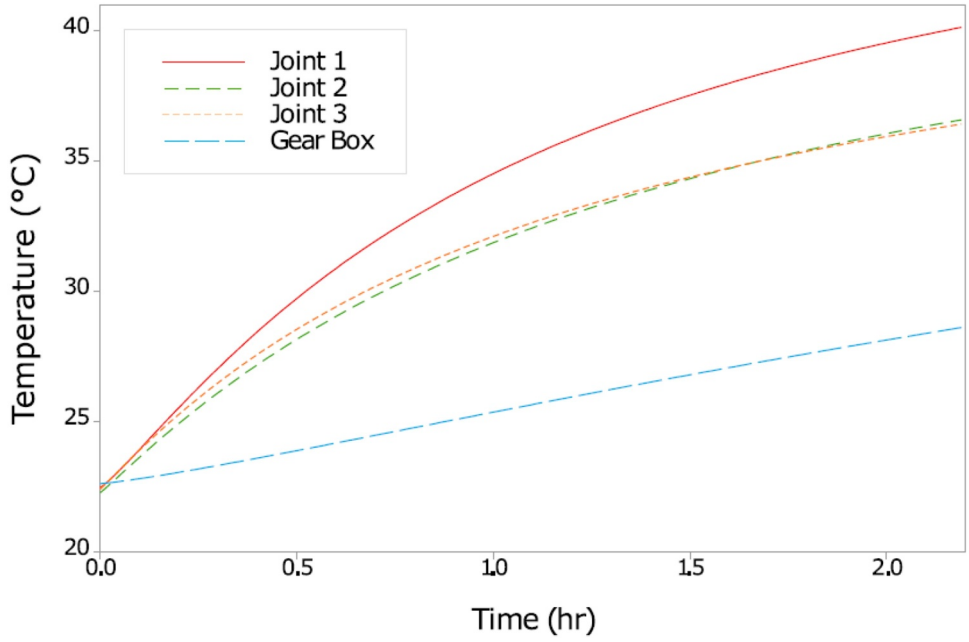



Figure 2. Temperature of gearbox and each joint during warm up. We find that motor casing temperatures vary greatly from gearbox casing temperature.

Table 1. $2^4$ design of experiments.

| Factors | Specification | Low Level (-) | High Level (+) |
|---|---|---|---|
| S | Linear Speed (%) | 25 | 75 |
| A | Acceleration (%) | 30 | 80 |
| L | Payload (kg) | 25 | 65 |
| T | J1 temperature (°C) | 22.5 | 40 |

motion, while carrying constant load, this is in order to maximise isolated movements to understand how each joint consumes energy and how each joint heats up.

The low values for speed and acceleration were chosen based on previous findings by Smetnova (2010) and Liu et al. (2018) as discussed in the Parameter Optimisation subsection, with high values chosen to be 50% higher in order to understand the relationship between these values on EC. The low value for payload was chosen to mimic an IR end effector of low weight such as a welding gun, while the high value was chosen to be 50% of the maximum payload of 130 kg, as IR used in manufacturing settings tend to be selected for their jobs with a factor of safety built-in. Finally, temperature values were chosen as room temperature in order to represent a robot's cold start, and the high value that an IR would achieve over constant operation over the course of a factory shift, which is towards its equilibrium temperature. These chosen temperatures are representative of operating temperatures that a robot might experience in the course of actual operation.

## 4 Results and analysis

Due to the way the DOE is set up, the 24 factorial experiments had 16 individual test cases, and with a second replicate, this resulted in a total of 32 individual runs. Replicates were obtained to have more reliable data, as the replicates allow us to minimise the impact of errors by averaging. Each set of experiments produced energy consumption data for each joint and produced additional cycle time information, all of which can be referenced in Table 2.

Table 2. Energy consumption and cycle time experimental data for each joint and replicate.

| Energy consumption (kWh) | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Run | Coded factors | | | | J1 | | J2 | | J3 | | Cycle times (s) | |
| Label | S | A | L | T | (r1) | (r2) | (r1) | (r2) | (r1) | (r2) | (r1) | (r2) |
| **1** | − | − | − | − | 9.75 | 10.18 | 34.04 | 33.56 | 40.52 | 41.03 | 125.2 | 125.3 |
| **S** | + | − | − | − | 1.78 | 1.89 | 4.68 | 4.67 | 5.15 | 5.13 | 45.3 | 45.5 |
| **A** | − | + | − | − | 3.41 | 3.58 | 10.63 | 10.32 | 11.91 | 11.98 | 69.2 | 69 |
| **SA** | + | + | − | − | 1.35 | 1.28 | 2.44 | 2.41 | 2.42 | 2.36 | 31.3 | 31.2 |
| **L** | − | − | + | − | 11 | 10.32 | 36.87 | 36.98 | 53.62 | 53.35 | 125 | 125 |
| **SL** | + | − | + | − | 1.81 | 1.81 | 5.18 | 5.12 | 6.83 | 6.9 | 45.5 | 45.4 |
| **AL** | − | + | + | − | 3.47 | 3.57 | 11.44 | 11.35 | 16.38 | 16.25 | 69.3 | 69.1 |
| **SAL** | + | + | + | − | 1.28 | 1.3 | 2.62 | 2.6 | 3.17 | 3.23 | 31.2 | 31.2 |
| **T** | − | − | − | + | 10.88 | 9.71 | 34.39 | 33.79 | 42.21 | 42.21 | 124.9 | 125 |
| **ST** | + | − | − | + | 1.8 | 1.77 | 4.74 | 4.77 | 5.36 | 5.38 | 45.3 | 45.5 |
| **AT** | − | + | − | + | 4.77 | 4.78 | 14.78 | 14.86 | 17.88 | 17.73 | 81.9 | 82.1 |
| **SAT** | + | + | − | + | 1.22 | 1.17 | 2.37 | 2.41 | 2.53 | 2.5 | 31.1 | 31.2 |
| **LT** | − | − | + | + | 10.13 | 10.14 | 37.83 | 37.52 | 55.72 | 55.4 | 125.1 | 125.1 |
| **SLT** | + | − | + | + | 1.72 | 1.78 | 5.15 | 5.13 | 7.05 | 7.19 | 45.3 | 45.3 |
| **ALT** | − | + | + | + | 4.88 | 4.93 | 16.1 | 16.16 | 23.41 | 23.54 | 81.9 | 82.1 |
| **SALT** | + | + | + | + | 1.31 | 1.18 | 2.65 | 2.64 | 3.34 | 3.24 | 31.4 | 31.2 |

### 4.1 EC significance of joints 1–3

According to the analysis, J1's sources of variations that contributed most to EC was linear speed (S), which contributed 62:61%, with the next highest contributions coming from acceleration (A) at 21.2% and the combination of linear speed and acceleration (SA) at 14.94%, as seen in Table 4. The rest of the factors and combinations contributed less than 2% to the EC. Payload (L) did not play any role in energy consumption for this joint.

Next, J2's greatest contribution came from linear speed (S), with a contribution percentage of 61.98%. The second greatest contributor was from the acceleration (A) factor at 22.31%. The third most significant contribution came from the combination of linear speed and acceleration (SA) at 14.48%. The rest of the factors make up less than 2% of the EC, which can be observed in Table 5.

Finally, the EC contribution for J3 was greatest from the linear speed factor which contributed 59.78%. The next most significant contributions came from the acceleration (A) and the combination of linear speed and acceleration (SA) which were 21.58% and 14.04%, respectively. Most interestingly, J3's other factors contributed more than 3%, with 1.99% being contributed by payload (L), as seen in Table 6.

In general, for all joints, linear speed (S) was found to have the greatest contribution, between 59.78% and 73.78%, to EC and then acceleration (A), found to contribute between 19.44% and 22.31%. After the contribution found from the combination of linear speed and acceleration (SA), which ranged from 5.99% to 14.94%, the next highest contributor varied. Cycle time information can be seen in Table 3, where the most significant contributor to cycle time is observed to be linear speed (S) at 73.78%.

Additionally, in Figure 3, we found that, regardless of cycle time and total power used, the power percentage distribution between the joints remained largely the same. This means any decrease of EC in any one joint reflects across the entire robot, so that it may be possible to monitor only one joint's EC to reliably reduce EC across the entire system while monitoring less data overall.

Table 3. ANOVA for cycle time (s).

| Source of variation | SS | % Cont |
|---|---|---|
| **S (linear speed)** | **30,773.1** | **73.78** |
| A (Acceleration) | 8106.7 | 19.44 |
| T (Temperature) | 80.3 | 0.19 |
| SA | 2497.6 | 5.99 |
| ST | 82.3 | 0.2 |
| AT | 84.8 | 0.2 |
| SAT | 82.9 | 0.2 |
| Error | 0.3 | 0 |

Table 4. ANOVA for Joint 1 EC (kWh).

| Source of Variation | SS | % Cont |
|---|---|---|
| **S (linear speed)** | **258.988** | **62.61** |
| A (Acceleration) | 87.703 | 21.2 |
| T (Temperature) | 0.602 | 0.15 |
| SA | 61.794 | 14.94 |
| ST | 0.933 | 0.23 |
| AT | 0.99 | 0.24 |
| SAT | 1.07 | 0.26 |
| Error | 1.593 | 0.39 |

Table 5. ANOVA for joint 2 EC (kWh).

| Source of variation | SS | % Cont |
|---|---|---|
| **S (linear speed)** | **3425** | **61.98** |
| A (Acceleration) | 1233.12 | 22.31 |
| L (Payload) | 13.13 | 0.24 |
| T (Temperature) | 12.97 | 0.23 |
| SA | 800.38 | 14.48 |
| SL | 7.29 | 0.13 |
| ST | 12.6 | 0.23 |
| AL | 2.99 | 0.05 |
| AT | 7.93 | 0.14 |
| SAL | 2.06 | 0.04 |
| SAT | 8.23 | 0.15 |
| Error | 0.6 | 0.01 |

Table 6. ANOVA joint 3 EC (kWh).

| Source of variation | SS | % Cont |
|---|---|---|
| **S (linear speed)** | **6366.8** | **59.78** |
| A (Acceleration) | 2298.2 | 21.58 |
| L (Payload) | 211.8 | 1.99 |
| T (Temperature) | 37.2 | 0.35 |
| SA | 1495.7 | 14.04 |
| SL | 120.4 | 1.13 |
| ST | 31.3 | 0.29 |
| AL | 40.1 | 0.38 |
| AT | 10.6 | 0.1 |
| SAL | 24.9 | 0.23 |
| SAT | 12 | 0.11 |
| Error | 0.4 | 0 |

### 4.2 Design of experiments analysis

In order to understand the results, we carried out a linear DOE analysis. The two-level design of the experiment assumes a linear relationship between the variables, but a 3+ level experiment could yield curvature in the factor space. We chose 5% in order to be conservative in our statistical analysis for significance. The analysis of variance (ANOVA) tables, seen in Table 3–6, provides information on how each parameter and combination of parameters contributes to either Cycle Time or EC for a given joint. Only factors with significant contribution are displayed, so we are able to see that first-order effects contributed most, with smaller contributions from second- and third-order interactions. As a result, other than the speed and acceleration (SA) combination, interactions between parameters were minimal.

Furthermore, as seen in Figure 4, we can observe the effects of low and high factors on EC in each joint. For J1, it was found that payload (L) did not contribute significantly to EC and was therefore not included. Simply by adjusting the speed and acceleration parameters from low to high, J1 can save about 10.5 kWh, while running at high temperatures will reduce savings by less than 1 kWh. In J2, the savings from speed and acceleration are larger, nearly 34 kWh from using a high speed versus a low speed. J2 will consume about an additional 2 kWh from increasing load and temperature. Finally, J3 saves the most with high speed and acceleration at a combined 48 kWh, but will lose about 7 kWh from increased load and temperature. Overall, the effective overall savings using the optimised response in Table 7 is 95 kWh, which represents a saving of about 44% from the least efficient case.

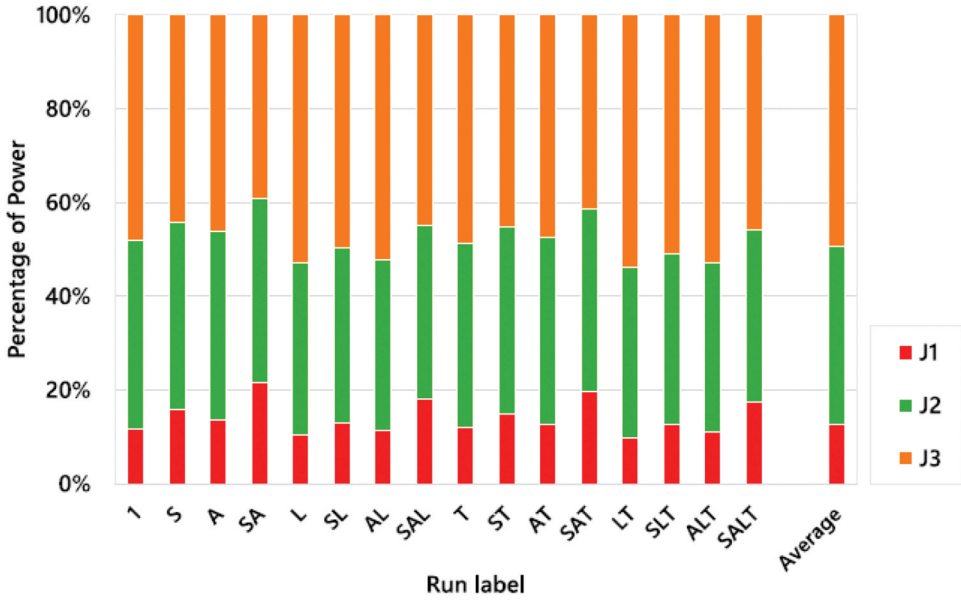



Figure 3. Percentage of energy consumption by joint.

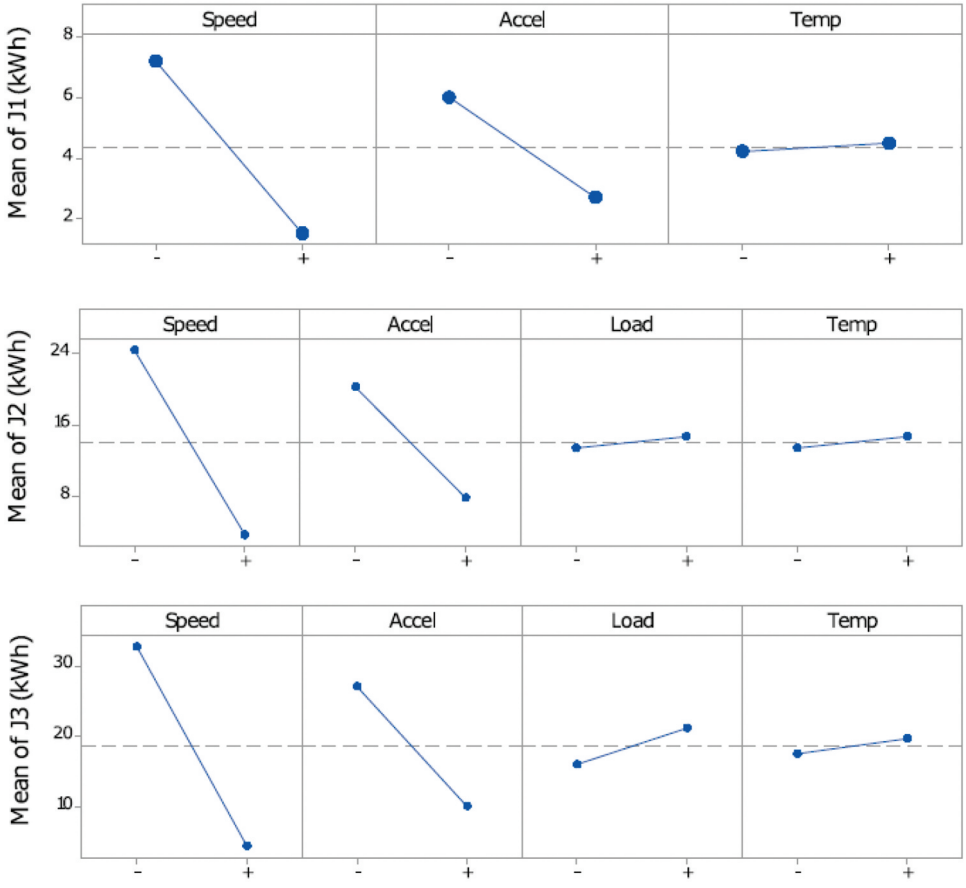



Figure 4. Main effects plots for each joint.

Table 7. Optimised response.

| S | A | L | T | Composite desirability |
|---|---|---|---|---|
| + | + | − | + | 0.986451 |

For all joints, low linear speeds (S) contributed to higher EC, with high linear speeds contributing to lower EC. In general, contributions from payload (L) and temperature (T) were

relatively small, but for both payload and temperature, levels corresponded directly to EC.

Figure 5 through 7 display representative power consumption from run 1 for each joint, chosen due to the relative prominence of its holding regions. Importantly, though the overall time for each run varied, all experiments followed the exact same programmed path, which is best illustrated by following J1, in Figure 5: the robot first rotates counterclockwise (CCW) to the first 'pick' point in the first 30 seconds, pauses for 10 seconds to 'pick up an object' before rotating CCW again to the 'place' point beginning near 55 seconds. Finally, the robot rotates clockwise (CW) to its 'home' position and repeats the path once with shorter holds between actions. It must be noted that, although the trajectory is representative of a pick and place operation, it is not a true pick and place operation, as the payload never changes through the trajectory as it is bolted onto the end effector. Towards this, the experiment best evaluates the sub-operation of transporting an object over a pick and place trajectory as opposed to actually evaluating a true pick and place operation.

### 4.3 Discussion of results

While the ANOVA results for each joint were largely similar, seen in Table 4–6, we need to address their differences. J1 saw that payload had no effect on its energy consumption, which likely originates from how the joint is built. Due to the need to swivel constantly, the joint sits on top large ball bearings which supports both the arm and payload mass near along with the rest of the robot designed to bear this load. With the Kawasaki's payload limit at 130 kg, the ball bearings and structure should be able to support that amount of mass with little effort. In Figure 5, it is easy to see that during holding periods, such as that between about 30 and 40 seconds, that the EC is nearly 0 kW. As a result, resting EC at J1 is low.

Referring back to Figure 2, although we did not measure the temperature of J1's gearbox, we can assume that, like J2's gearbox, the temperature did not increase significantly over the warm up period. The temperature of J1's actual gears is likely to be even lower than the recorded temperatures for J2 GB, because its housing and volume of grease are even greater than that of J2.

Payload contributions to J2 are low due to the counterbalancing mechanisms built in to the Kawasaki. This allows J2 to operate relatively independently of the payload, though not as much as J1. As seen in Figure 6, resting EC, as seen between 40 and 50 seconds, J2 has a resting EC of about 1 kW. Temperature of J2 has a slight contribution to increasing energy consumption likely due to temperature effects on motor current draw. From the measurement of J2 GB, it is unlikely that the temperature of the actual gears would heat up significantly from general usage over a factory work day. Assuming that J2 GB would heat up similarly to that of the motor casing, such that it should reach some equilibrium point, it would take between 12 and 14 hours for the gearbox to reach an equilibrium temperature, but only under constant usage and high load as seen in Equation 1, which was found through regression analysis, performed on the data seen in Figure 2. Presumably, most of the heat within the gearbox is actually generated by the motors and transferred via conduction, as the volume of grease is very high and the friction within the gears relatively low.

$$T_{J2GB} = 36.4719 + (-13.8537)\, e^{-0.2207 t^{\wedge}(1.2011)} \quad (1)$$

J3's overall contribution to power usage tends to be highest, as seen in Figure 3, with its holding EC value, as seen between 40 and 50 seconds in Figure 7, to be a little under 1 kW. The reason for J3's percentage of power being the highest of the three joints is likely due to the nature of how the IR is built, since this joint must support the entirety of the arm and payload, the IR consumes energy to overcome the moment generated by the payload and arm mass.

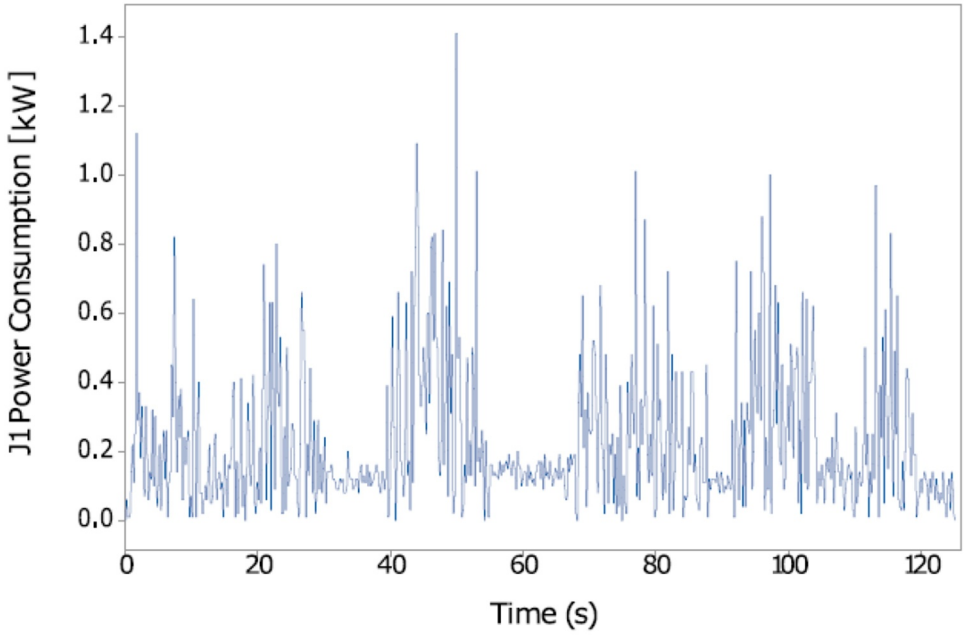



Figure 5. Representative power consumption for J1. Data from run 1.

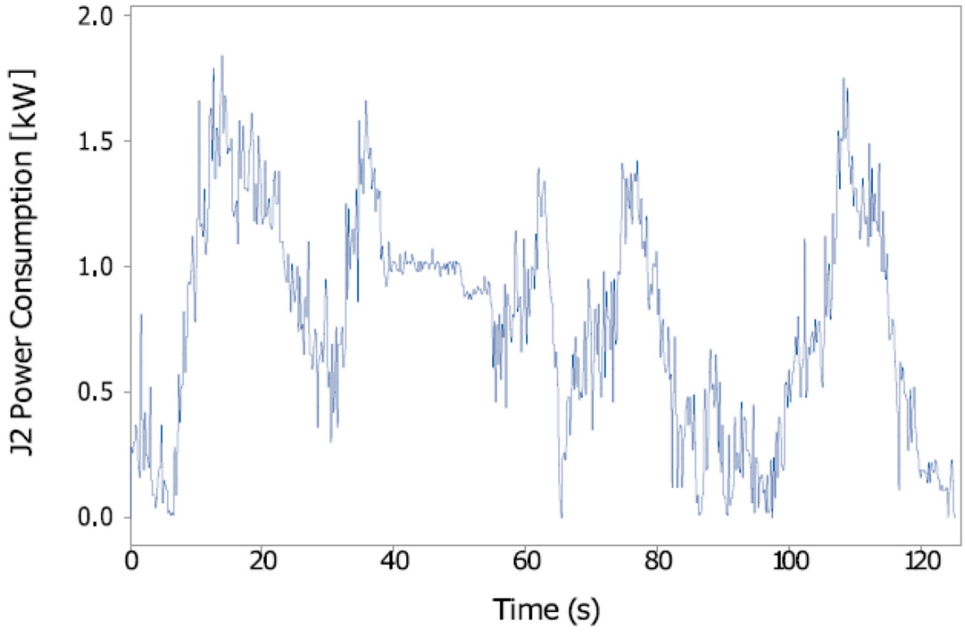



Figure 6. Representative power consumption for J2. Data from run 1.

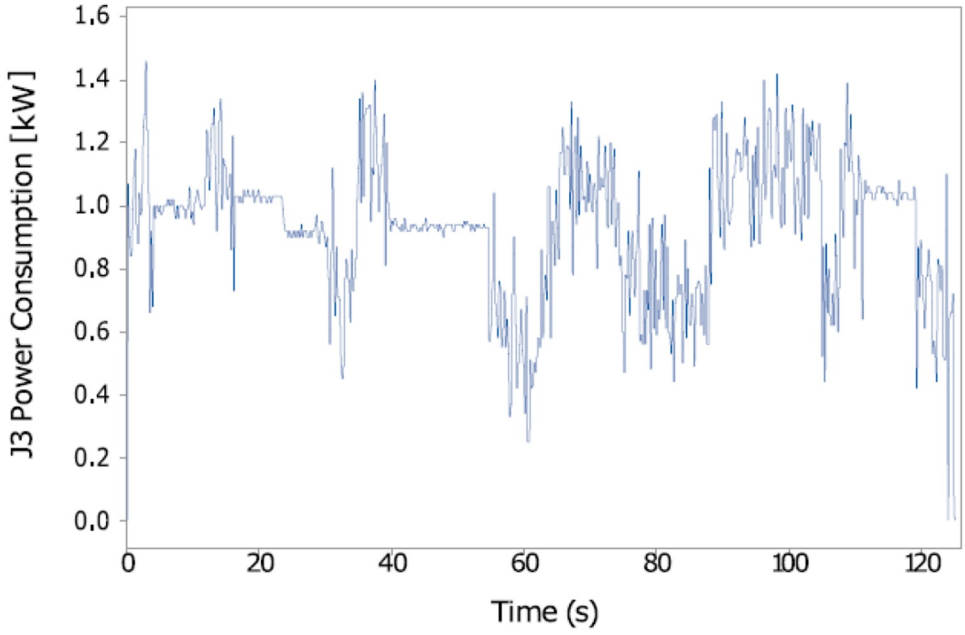



Figure 7. Representative power consumption for J3. Data from run 1.

Naturally, speed and acceleration will also contribute to changes in cycle time. For the Kawasaki ZZX130L, as seen in Figure 8, there is a positive relationship between faster movements and lower EC and between slower movements and higher EC. However, based on this figure and the linear equations, there is cycle time near 60 seconds where it is more beneficial to use lower speed and acceleration, as shown by where the linear equations are equivalent. It is also important to note that J2 and J3 behave similarly to the presented results. As a result, depending on the existing cycle time of a process, EC can be lowered by choosing either the high or low factor levels.

However, in general, as found by response optimisation in Minitab, as seen in Table 7, using a high speed and acceleration tends to result in lower EC for this model of robot. Low load is suggested to minimise EC in joints 2 and 3, and due to the small contribution by the temperature factor, a high temperature or extended running time is suggested.

## 5 Conclusions and future work

An energy consumption analysis of linear speed, acceleration, payload, and temperature was conducted in this study using design of experiment methodology to evaluate which parameter contributes most to EC. Current literature is focused in motion planning, scheduling, and parameter optimisation, often

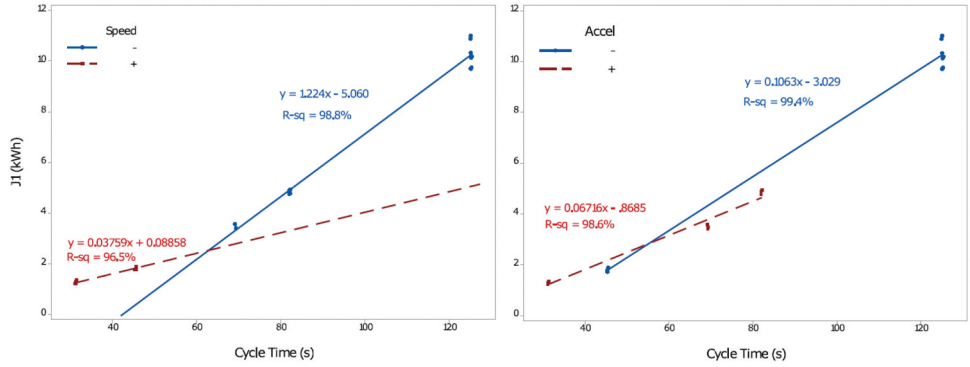



Figure 8. Regression trends for J1 EC values, sorted by high and low linear speed (S) and acceleration (A) factors.

without knowing which parameter in their experiment contributes most to their EC. Through the DOE analysis, we are able to conclude that both linear speed (S), acceleration (A), and their combination (SA) contribute an upwards of 95% to EC in the observed joints. The fact that linear speed and acceleration interact so heavily is not surprising, given that acceleration is simply the derivative of velocity.

Our research effectively finds and improves on the understanding of how speed and acceleration contribute to EC, which can now be used to explain the effectiveness of research in the motion planning regime, like the works of Paryanto, Bornschlegl, and Franke (2015) and Mohammed et al. (2014). Our findings also agree with those of Chemnitz, Schreck, and Krüger (2011) in that there may be particular processes where the IR should move quickly and idle or move slowly to reach its target point, though we are now able to suggest that it may also be a cycle time issue, as opposed to a issue related to robot model or age. In particular, our findings suggest this depends on overall cycle time, as shown in Figure 8. For cycles that are shorter, slower IR speeds and accelerations can be used to save energy, but as cycle times grow longer, faster movements are preferred.

Naturally, speed is also the greatest contributor to cycle time if a cycle has no set timing. Additionally, load does not affect cycle time, as the IR are designed to have high rates of repeatability, which naturally preclude this sort of variance based on payload. Future works should explore in more detail the speed curve of the robot's trajectory to energy consumed and validate the assumption that higher percentages of payload weight do not greatly affect energy usage.

Temperature effects, similar to those researched by (Eggers et al. 2019), are shown to have small contributions to EC, though our approach to measuring both motor casing and gearbox temperature allows us to conclude that, within the span of our warm-up exercise, the gearbox temperature does not greatly increase and cannot be easily related to motor casing temperature, whereas previous works have suggested that motor casing temperature was a good, relative measure of IR temperature. The minor effects of temperature on EC, as shown in Table 5 through 7 and in Figure 4, are likely due to temperature effects on motor currents, causing increased resistance within the servo motors. Most interestingly, though there is no payload effect on cycle time, there is a minor temperature effect on cycle, as attributed to the same motor changes as before. To better understand if load has the potential to affect cycle time, pursuing an additional experiment with more extreme payloads is suggested.

Although the current study has established a strong relationship between speed, acceleration, and EC, this research can be further improved by experimenting with other levels of speed, acceleration, payload, and temperature, which will help determine the validity of the assumption of linearity. Another line of possible experimentation should be to consider different process types like welding or palletising exercises or to investigate a robot's age and wear as it relates to energy consumption changes over its lifetime. Finally, given the framework in this paper, using DOE is a robust method to understanding the EC of a particular type of robot in relationship to certain factors and should be used to further determine if EC across models and ages of IR are similar or vary greatly by manufacturer or other variables.

## Disclosure statement

No potential conflict of interest was reported by the authors.

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